

Lab files

Sunday, April 26, 2026 1:02 PM

[SOC101/course_files/04_SIEM.zip at main · MalwareCube/SOC101 · GitHub](#)

has all the examples , tools , and challenges

File password :

LCty2JmHQLB3uc9VxjeohENr

Introduction to SIEM and Log Management

Sunday, June 29, 2025

1:37 PM

- **Log Management**
- **Real-time Monitoring**
- **Alerting and Notification**
- **Incident Response**
- **Dashboards, Reports, and Visualization**
- **Threat Intelligence Integration**

● Collection	● Which devices will we collect events from? ● Which events do we want to collect? ● How long will we retain the logs? ● Where will we store the logs? ● Methods of collection - Agents - Agentless
● Aggregation	● Collecting and consolidating events ● Unify the timeline across the organization ● Enhancing holistic visibility

● Parsing and Normalization	● Ensure consistency across logs ● Extracting structured information ○ Fields, columns ○ Regular expressions, parsing tools, or custom parsers ● Convert into a common schema ○ Field Mapping ○ Data Transformation ○ Common Event Format (CEF)
● Retention	● Storing log data to ensure analysis availability ○ Incident Response ○ Compliance ● Log Retention Policies ○ Retention Period ○ Storage Solutions ○ Security Controls ○ Storage Integrity ○ Data Destruction
● Indexing	● Turns raw logs into searchable event data ● Repository or grouping of events ● Efficient log retrieval ● Helps with scaling as log sources grow
● Correlation and Analysis	● Linking related log events together ● Contextualization ○ Enriching events with additional metadata ● Correlation Rules ○ Specify how events should be correlated ● Correlation Engines

● Correlation and Analysis	● Linking related log events together ● Contextualization ○ Enriching events with additional metadata ● Correlation Rules ○ Specify how events should be correlated ● Correlation Engines ● Analysis ○ Pattern Recognition ○ Anomaly Detection
● Alerting	● Notify relevant people about security incidents ● Threshold-Based Alerts ● Pattern-Based Alerts ● Anomaly-Based Alerts ● Event-Based Alerts

SIEM Deployment Models

Thursday, August 14, 2025 12:51 AM

SIEM Components



Endpoints : creates log

Forwarder : take the logs and store them in database or storage area

Indexer : the database or storage area of logs

Search head : the siem gui where the analyst search and ask for data or logs that are brought from indexers

SIEM Deployment Models

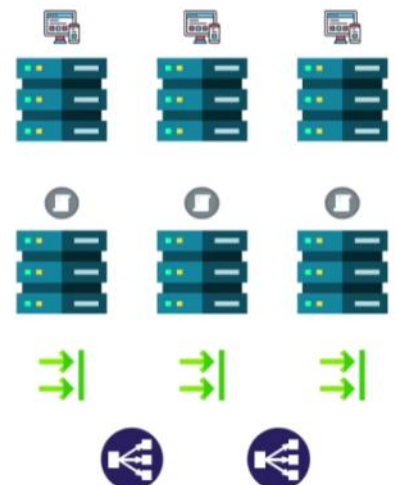
Single Instance Deployment



Distributed Deployment



Clustered Deployment



All components are in same server or machine	The components are divided on multiple machines or servers	Each component has its own place and server and this is usually used in big companies but high cost
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Log Types

Thursday, August 14, 2025 1:05 AM

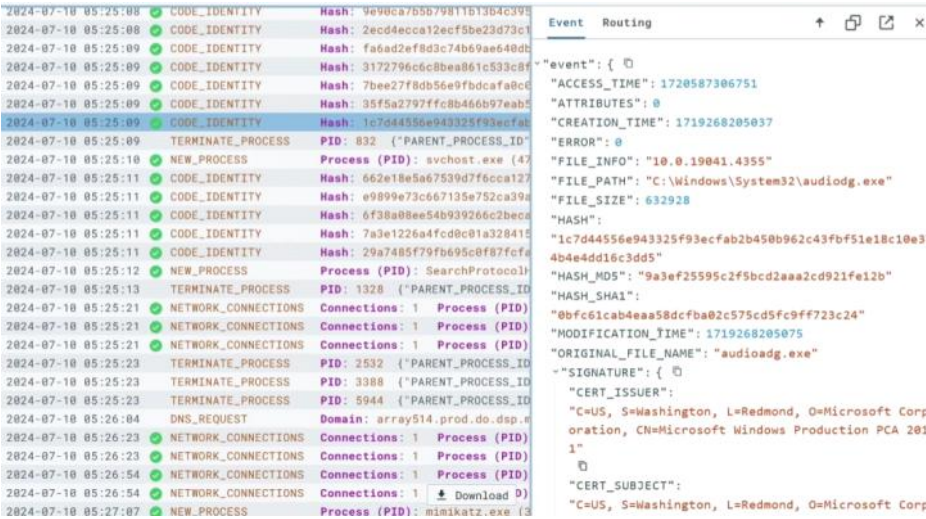
a **log** is a **record of events** or activities that occur in a computer system, network, or application.

● System Logs ○ Windows Event Logs ○ Sysmon Logs ○ Linux/Unix Syslogs	Example of a log the high light is a last whoami command <pre>tcm@SOC101-ubuntu:/var/log\$ tail auth.log 2024-07-16T20:22:21.672172-04:00 SOC101-ubuntu systemd-logind[576]: Session c1 logged out. Waiting for processes to exit. 2024-07-16T20:22:21.749180-04:00 SOC101-ubuntu polkitd[559]: Unregistered Authentication Agent for unix-session: c1 (system bus name :1.39, object path /org/freedesktop/PolicyKit1/AuthenticationAgent, locale en_US.UTF-8) (disconnected from bus) 2024-07-16T20:22:21.778482-04:00 SOC101-ubuntu systemd-logind[576]: Removed session c1. 2024-07-16T20:25:01.359272-04:00 SOC101-ubuntu CRON[2978]: pam_unix(cron:session): session opened for user tcm(uid=1000) by tcm(uid=0) 2024-07-16T20:25:01.368505-04:00 SOC101-ubuntu CRON[2977]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0) 2024-07-16T20:25:01.382032-04:00 SOC101-ubuntu CRON[2977]: pam_unix(cron:session): session closed for user root 2024-07-16T20:25:46.392526-04:00 SOC101-ubuntu CRON[2978]: pam_unix(cron:session): session closed for user tcm 2024-07-16T20:26:56.857203-04:00 SOC101-ubuntu sudo: tcm : TTY=pts/1 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/whoami 2024-07-16T20:26:56.861785-04:00 SOC101-ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by tcm(uid=1000) 2024-07-16T20:26:56.867315-04:00 SOC101-ubuntu sudo: pam_unix(sudo:session): session closed for user root</pre>
● Network Logs ○ Firewall Logs ○ Proxy Logs ○ DNS Logs	track all network traffic and behavior across systems. Includes: • IP addresses (source/destination) • Ports used • Protocols (TCP, UDP) • Allowed/blocked traffic • Bandwidth usage
● Application Logs ○ Database Logs ○ Web Server / HTTP Logs ○ App Logs	Apache log <pre>200.96.104.241 - - [12/Sep/2023:09:44:28 -0300] "GET /modules.php?name=Downloads&&%20lid=-1%20UNION%20SELECT%20, user name, user_id, user_password, name, %20user_email, user_level, 0, 0%20FROM%20admin_users HTTP/1.1" 200 9918 "-" "Mozilla/4.0"</pre>
● Security Logs ○ Authentication Logs ○ IDS/IPS Logs ○ Endpoint Security Logs	To log events related to security controls and threats. Includes: • Malware detections • Failed login attempts • Unauthorized access • Security software actions • Suspicious behavior
● Cloud Logs ○ AWS CloudTrail Logs ○ Azure Activity Logs / Log Analytics	To monitor activities in cloud environments (e.g., AWS, Azure, Google Cloud). Includes: • API calls • Resource provisioning/deletion • User access and actions • Service usage
● Audit Logs ○ Audit Trail Logs	track <i>who did what, when, and where</i> in a system for accountability and compliance. Includes: • User login/logout • File access/modification • Permission changes • Admin actions

Log Formats

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Sas

● Unstructured Logs ○ No predefined format or syntax ○ Common Log Format (CLF)	Similar to webserver log have the data we need but in unstructured wat <pre>192.168.1.1 - - [16/Jul/2024:10:15:32 +0000] "GET /index.html HTTP/1.1" 200 349 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/99.0.9999.99 Safari/537.36"</pre> 
● Semi-Structured Logs ○ Some syntax structure ○ Lack adherence to a schema ○ Syslog ○ Windows Event Log (EVT)	Logs that look as text with no schema Windows log example 
● Structured Logs ○ Well-defined syntax and formatting ○ Adherence to an agreed upon schema ○ CSV, TSV ○ JavaScript Object Notation (JSON) ○ eXtensible Markup Language (XML)	Used by the siem 

User Behavior Indicators

- **Multiple Failed Login Attempts**
 - Incorrect usernames or passwords
 - Increase in failures from a single user account
 - Increase in failures from multiple user accounts
- **Login Times**
 - Time of day that logons or access requests are taking place
 - Abnormalities from a user's baseline
- **Login/Access Locations**
 - Geographic locations of logons or access requests
 - Unusual countries or regions
 - Impossible Travel
- **File Access Patterns**
 - File paths, modifications, or other activity
- **User-Agent Strings**
 - Unusual or associated with known tools

SQL Injection

- Inserting or injecting malicious SQL statements
- Manipulate expected database queries
 - Retrieve sensitive information
 - Extract hashed passwords
 - Bypass authentication logic
- Look for SQL keywords
 - `SELECT, FROM, WHERE, UNION`

Data Queries

- **SELECT**
 - Retrieve data from a database table
- **FROM**
 - Specifies the table to retrieve data from
- **WHERE**
 - Filter records based on conditions
- **ORDER BY**
 - Sort the results (ascending or descending)
- **GROUP BY**
 - Group rows with identical values into summaries
- **DELETE**
 - Removes existing rows from a table
- **DROP**
 - Deletes a table or database from the system
- **JOIN**
 - Combines rows from two or more tables based on a column
- **INNER JOIN**
 - Returns records that have matching values in multiple tables
- **UNION**
 - Combines the results of two or more SELECT statements

Data Manipulation

- **INSERT INTO**
 - Adds new rows of data into a table
- **UPDATE**
 - Modifies existing data in a table

- Look for injection characters
 - Single/double quotes, semicolons, comment indicators (dashes)
- URL-encoded keywords/injection characters

SQL Injection Characters

Character	Normal Character	URL-Encoded Character
Single Quote	'	%27
Double Quote	"	%22
Space		%20
Semicolon	;	%3B
Dash	-	%2D

- **Malformed entries or errors within database logs**

Example

```
192.168.1.100 - - [15/Jul/2024:14:55:23 +0000] "GET /index.php?id=1%20OR%201=1;-- HTTP/1.1" 200 2326 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36"
```

```
"GET /index.php?id=1%20OR%201=1;--"
```

```
SELECT * FROM users WHERE id = 1;
```

```
SELECT * FROM users WHERE id = 1 OR 1=1; --
```

Seeing the 1=1 this showing sqli

You can also see SQLi payloads : <https://github.com/swisskyrepo/PayloadsAllTheThings/tree/master/SQL%20Injection>

Cross-Site Scripting

- **Executing malicious code by injecting JavaScript**
 - Hijack user sessions
 - Steal cookies
 - Deface websites
- **Look for `<script>` tag indicators**
- **Look for event handlers**
 - `onload, onclick, onmouseover`
 - References to `<script>` tags
 - References to `"javascript"`
- **Special characters**
 - `<, >, ", ', &, %, ;`
- **URL-encoded injection characters**

XSS examples

```
192.168.1.100 - - [15/Jul/2024:14:55:23 +0000] "GET
/search?q=<script>alert('XSS')</script> HTTP/1.1" 200 2326 "-"
"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36"
```



```
192.168.1.101 - - [15/Jul/2024:14:56:45 +0000] "GET
/profile?name=%3Cimg%20src%3Dx%20onerror%3Dalert('XSS')%3E
HTTP/1.1" 200 2326 "-" "Mozilla/5.0 (Windows NT 10.0; Win64;
x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124
Safari/537.36"
```

```
<img src=x onerror=alert('XSS')>
```

You can also check other XSS payload in : <https://github.com/payloadbox/xss-payload-list>

Common Attack Signatures: Command Injection

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Command Injection

- Executing arbitrary OS commands
- Look for special characters that separate commands
 - ; , || , &&

Example of command injection :
using semi colon can make the usage of multiple commands

```
tcm@SOC101-ubuntu:~$ ls; cat /etc/passwd
Desktop    Downloads  msfinstall Pictures    snap      Templates
Documents  malware    Music      Public     SOC101    Videos
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
```

Using || will make the first command run if it has error the 2nd will run

```
tcm@SOC101-ubuntu:~$ ls || cat /etc/passwd
Desktop    Downloads  msfinstall Pictures  snap    Templates
Documents  malware    Music      Public    SOC101  Videos
tcm@SOC101-ubuntu:~$ ls ^/notexist || cat /etc/passwd
ls: cannot access '/notexist': No such file or directory
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
```

&& will make the 2nd work if the first is working

```
tcm@SOC101-ubuntu:~$ ls && cat /etc/passwd
Desktop    Downloads  msfinstall Pictures  snap    Templates
Documents  malware    Music      Public    SOC101  Videos
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
```


- **Look for references to commands or utilities**
 - `ls, echo, bash, cat, cd, cmd.exe, curl, wget`
 - `ping, sudo, chmod, rm, nc, nc.exe, sh`
- **URL-encoded injection characters**

```
192.168.1.103 - - [15/Jul/2024:15:30:00 +0000] "GET  
/execute?cmd=ls;cat%20/etc/passwd HTTP/1.1" 200 4123 "-"  
"Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36  
(KHTML, like Gecko) Chrome/91.0.4472.124 Safari/537.36"
```

Common Attack Signatures: Path Traversal and Local File Inclusion

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Path Traversal / Local File Inclusion

- **Accessing files outside of the web root**
 - Include or execute unintended files/scripts
- **Path Traversal**
 - Access files/directories outside of the web root
 - Enumerate the system, read hardcoded credentials
- **Local File Inclusion (LFI)**
 - Include a local file from the system
 - Enumerate the system, read hardcoded credentials
 - Execute scripts / remote code execution
- **Look for path traversal sequences**
 - URL-encoded characters



```
192.168.1.111 - - [15/Jul/2024:16:30:00 +0000] "GET  
/download?file=%2e%2e%2f%2e%2e%2f%2e%2e%2fetc%2fpasswd HTTP/1.1"  
200 3182 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64)  
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/91.0.4472.124  
Safari/537.36"
```

file=../../../../etc/passwd

Command Line Log Analysis

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We are making log analysis manually

```
tcm@SOC101-ubuntu:~$ cd Desktop/04_Siem/01_Log_Analysis/  
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ ls  
access.log  events.json  json_alert.sh
```

Step 1 : know what is in it

file -> this command lets u know the file and what u may expect in it

```
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ file access.log  
access.log: ASCII text
```

Step 2 : know the size and permissions

Ls -lh -> will show the permissions and the file size and owner

```
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ ls -lh access.log  
-rw-rw-r-- 1 tcm tcm 343K Jul 18 12:22 access.log
```

Note: -h will make the size more readable

Step 3 : know number of lines and words and chars

Wc -> is used to show number on lines - words - bytes (characters)

```
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ ls -lh access.log  
-rw-rw-r-- 1 tcm tcm 343K Jul 18 12:22 access.log  
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ wc access.log  
 3086  41767 350810 access.log  
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ wc -l access.log  
3086 access.log  
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ wc -w access.log  
41767 access.log  
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ wc -c access.log  
350810 access.log
```

Wc -l -> for lines only

Wc -w -> for words

Wc -c -> for characters or bytes only

Step 4 : know the time

Head -> first lines (if u added -n 1 -> first line of the file)

Tail -> last lines (if u added -n 1 ->last line of the file)

```
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ head access.log -n 1
91.15.183.209 - - [17/Jul/2024:18:47:37 -0400] "GET /users/02492184 HTTP/1.1" 200 200 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.0.0 Safari/537.36"
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ tail access.log -n 1
82.35.67.148 - - [17/Jul/2024:18:50:13 -0400] "GET /users/492849 HTTP/1.1" 200 200 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.0.0 Safari/537.36"
```

Step 4: extract same data info

Here we want to get the same data type and use it alone instead of having all same time

For example we want the ip address only so we can use

Cut -d " " -f 1

-d "": in the side the quote we add the sign to cut info like space ; " " :

-f : here which part example 1 -> ip 7-> the url path is requested

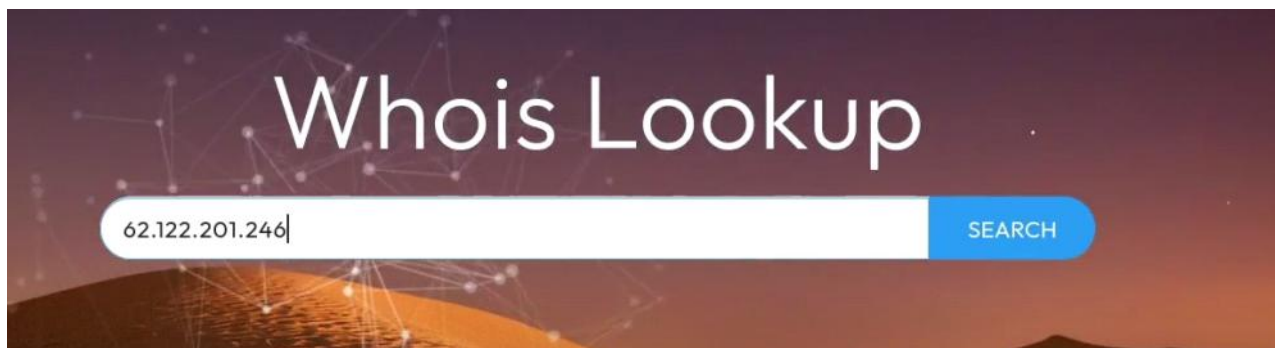
```
91.15.183.209 - - [17/Jul/2024:18:47:37 -0400] "GET /users/02492184 HTTP/1.1" 200 200 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.0.0 Safari/537.36"
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ cut access.log -d " " -f 1
```

If u add sort and uniq -> will remove replicates and also sort them

Add -c : will count the number of times of repetition (this could make an ip for example suspicious and search about it)

And using sort -nr will sort them according and make the most used at the top for easy

```
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ cut access.log -d " " -f 1 | sort | uniq -c | sort -nr
2920 62.122.201.246
7 108.175.99.111
1 99.99.241.222
1 98.128.157.54
1 95.69.14.49
1 94.38.121.227
1 9.187.26.142
```



Or use cisco talos

OWNER DETAILS

DOMAIN

m8a8a8y.site

CONTENT DETAILS

CONTENT CATEGORY

Computer Security

Think these category details are incorrect?

Submit Content Categorization Ticket

REPUTATION DETAILS

WEB REPUTATION

Neutral

Submit Web Reputation Ticket

BLOCK LISTS

TALOS SECURITY INTELLIGENCE BLOCK LIST

ADDED TO BLOCK LIST

No

And we can do the something to user agent

```
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ cut -d "\"" -f 6 access.log | sort | uniq -c | sort -nr
  2891 Mozilla/5.0 (Hydra)
    45 Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:127.0) Gecko/20100101 Firefox/127.0
    34 Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/125.0.0.0 Safari
/537.36
    31 Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.0.0 Safari
/537.36
    29 Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
    28 Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.0.0
Safari/537.36
    24 Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/126.0.0.0 Safari
/537.36 Edg/126.0.0.0
    4 -
```

Look at the green and yellow highlight

The yellow : if we google it this seems as the hydra tool

"Mozilla/5.0 (Hydra)"

Information Security Stack Exchange

https://security.stackexchange.com › questions › hydra-...

Hydra bruteforce and JSON

May 13, 2014 — if (strcmp(type, "POST") == 0) { sprintf(buffer, "POST %.600s HTTP/1.0\r\nHost: %s\r\nUser-Agent: Mozilla/5.0 (Hydra)\r\nContent-Type: ...

The green : it seems like there is a nmap scan doing

And then we can make a grep to see what is it doing

For example its trying to brute force the login.php

```
grep "Mozilla/5.0 (Hydra)" access.log
```

```

62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "POST /login.php HTTP/1.0" 200 312 "-" "Mozilla/5.0 (Hydra)"
62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "GET /login.php HTTP/1.0" 200 801 "-" "Mozilla/5.0 (Hydra)"
62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "GET /admin.php HTTP/1.0" 200 476 "-" "Mozilla/5.0 (Hydra)"
62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "POST /login.php HTTP/1.0" 200 312 "-" "Mozilla/5.0 (Hydra)"
62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "GET /login.php HTTP/1.0" 200 801 "-" "Mozilla/5.0 (Hydra)"
62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "POST /login.php HTTP/1.0" 200 312 "-" "Mozilla/5.0 (Hydra)"
62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "GET /login.php HTTP/1.0" 200 801 "-" "Mozilla/5.0 (Hydra)"
62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "POST /login.php HTTP/1.0" 200 312 "-" "Mozilla/5.0 (Hydra)"
62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "GET /login.php HTTP/1.0" 200 801 "-" "Mozilla/5.0 (Hydra)"

```

And we see the ip that is using it

Some times when make a login and the credentials are wrong the response is 200 but u may want to look for redirection
Use ok like cut

Example : `awk '{print $9}'` will print response time

```
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ grep "Mozilla/5.0 (Hydra)" access.log | awk '{print $9}'
```

```

200
200
200
200
200
200
200
200
200
200
200
200

```

Here we used it to return the repos that are more than 200

```
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ grep "Mozilla/5.0 (Hydra)" access.log | awk '$9 > 200'
62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "POST /login.php HTTP/1.0" 302 301 "-" "Mozilla/5.0 (Hydra)"
```

Also we can filter on the same time it happens and see if there is another page the attacker my login in

```
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ grep "17/Jul/2024:18:49:02 -0400" access.log | grep -v "login.php"
62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "GET /admin.php HTTP/1.0" 200 476 "-" "Mozilla/5.0 (Hydra)"
```



Log Analysis Challenge 1

Saturday, August 16, 2025 5:55 PM

<https://challenges.malwarecube.com/#/c/d63f1671-a312-4677-8684-3cf446870536>

Pattern Matching

Monday, April 20, 2026 10:26 AM

Here we are using grep which is used to get the match on entered string

1. Simple use

```
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ grep "404" access.log
62.122.201.246 - - [17/Jul/2024:18:48:13 -0400] "GET /robots.txt HTTP/1.1" 404 451 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
62.122.201.246 - - [17/Jul/2024:18:48:13 -0400] "POST /sdk HTTP/1.1" 404 451 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
62.122.201.246 - - [17/Jul/2024:18:48:13 -0400] "GET /nmaplowercheck1721256492 HTTP/1.1" 404 451 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
62.122.201.246 - - [17/Jul/2024:18:48:13 -0400] "GET /.git/HEAD HTTP/1.1" 404 451 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
62.122.201.246 - - [17/Jul/2024:18:48:13 -0400] "GET /evox/about HTTP/1.1" 404 451 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
62.122.201.246 - - [17/Jul/2024:18:48:13 -0400] "GET /HNAP1 HTTP/1.1" 404 451 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
62.122.201.246 - - [17/Jul/2024:18:48:13 -0400] "GET /favicon.ico HTTP/1.1" 404 451 "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)"
108.175.99.111 - - [17/Jul/2024:18:49:29 -0400] "GET /admin/config.php HTTP/1.1" 404 432 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:127.0) Gecko/20100101 Firefox/127.0"
```

2. Grep count :

grep -c : is used to count the number of matches found

```
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ grep -c "404" access.log
8
```

3. Expression match:

grep -E : is used to find the expression match , and here u can use the normal ones as > or url encoded %3E

U can use it to get the XSS or SQLinj for example and other attacks forms

```
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ grep -E '%3C|%3E|<|>' access.log
108.175.99.111 - - [17/Jul/2024:18:49:23 -0400] "GET /search.php?q=%3Cscript%3Ealert%28%29%3C%2Fscript%3E HTTP/1.1" 200 147 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:127.0) Gecko/20100101 Firefox/127.0"
tcm@SOC101-ubuntu:~/Desktop/04_SIEM/01_Log_Analysis$ grep -E '\.\/|/%2E%2E%2F|/%2E%2E%2E%2F%2F' access.log
108.175.99.111 - - [17/Jul/2024:18:49:41 -0400] "GET /stats.php?page=../../../../ HTTP/1.1" 200 147 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:127.0) Gecko/20100101 Firefox/127.0"
108.175.99.111 - - [17/Jul/2024:18:49:45 -0400] "GET /stats.php?page=../../../../robots.txt HTTP/1.1" 200 147 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:127.0) Gecko/20100101 Firefox/127.0"
108.175.99.111 - - [17/Jul/2024:18:49:49 -0400] "GET /stats.php?page=%2E%2E%2F%2E%2E%2F%2E%2E%2F%2E%2E%2Fetc%2Fpasswd HTTP/1.1" 200 3073 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:127.0) Gecko/20100101 Firefox/127.0"
108.175.99.111 - - [17/Jul/2024:18:49:53 -0400] "GET /stats.php?page=%2E%2E%2F%2E%2E%2F%2E%2E%2F%2E%2E%2Fetc%2Fshadow HTTP/1.1" 200 147 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:127.0) Gecko/20100101 Firefox/127.0"
```

And as u see all http-code:reponse-size code are 200 147 but one is 200 3073 which means the attacker has been logged in or found more info

Structured Log Analysis

Monday, April 20, 2026 10:42 AM

1. If u are looking for logs in file of form of JSON use : jq . Filename.json

```
{
  "PARENT_PROCESS_ID": 1948,
  "PROCESS_ID": 3532,
  "THREADS": 2,
  "USER_NAME": "DESKTOP-C60J5NL\\tcm"
},
{
  "routing": {
    "arch": 2,
    "did": "",
    "event_id": "9f34c375-8841-4d01-b392-1e760072dd3a",
    "event_time": 1720589227286,
    "event_type": "NEW_PROCESS",
    "ext_ip": "192.168.1.9",
    "hostname": "desktop-c60j5nl.net.example.com",
    "iid": "f02a8078-5960-4db0-9dd8-69fcac1e1df9",
    "int_ip": "192.168.1.9",
    "moduleid": 2,
    "oid": "1b6c5df7-ef3b-4521-bad6-f1f0ac9c13aa",
    "parent": "aae2fd84c3b0dd864492570c668e1b34",
    "plat": 268435456,
    "sid": "00b9df07-c52f-4140-a193-1405679ee5c9",
    "tags": [
      "windows",
      "workstations"
    ],
    "this": "1f89cc8078888206eac0c23e668e1bab"
  },
  "ts": "2024-07-10 05:27:07"
}
```

2. To see how many objects or events : jq 'length' filename.json

```
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ jq 'length' events.json
9
```

3. We select an event based on the ID or PID or File path or any other : jq '[] | select(.event.PROCESS_ID == 3532)' filename.json

```
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ jq '[] | select(.event.PROCESS_ID == 3532)' events.json
{
  "event": {
    "BASE_ADDRESS": 140695972675584,
    "COMMAND_LINE": "minikatz.exe",
    "FILE_IS_SIGNED": 1,
    "FILE_PATH": "C:\\Users\\tcm\\Downloads\\minikatz.exe",
    "HASH": "92804faaab2175dc501d73e814663058c78c0a042675a8937266357bcfb96c50",
    "MEMORY_USAGE": 9322496,
    "PARENT": {
      "BASE_ADDRESS": 140699238662144,
      "COMMAND_LINE": "\"C:\\Windows\\system32\\cmd.exe\"",
      "FILE_IS_SIGNED": 1,
      "FILE_PATH": "C:\\Windows\\system32\\cmd.exe",
      "HASH": "badf4752413cb0cbdc03fb95820ca167f0cdc63b597ccdb5ef4311180e008b0",
      "MEMORY_USAGE": 4354048,
      "PARENT_ATOM": "223290f2013a19ab33f84049668e1743",
      "PARENT_PROCESS_ID": 7020,
      "PROCESS_ID": 1948,
      "THIS_ATOM": "aae2fd84c3b0dd864492570c668e1b34",
      "THREADS": 2,
      "TIMESTAMP": 1720589108724,
      "USER_NAME": "DESKTOP-C60J5NL\\tcm"
    },
    "PARENT_PROCESS_ID": 1948,
    "PROCESS_ID": 3532,
    "THREADS": 2,
    "USER_NAME": "DESKTOP-C60J5NL\\tcm"
  },
  "event_id": "f"
}
```

Then u can pipe a specific event as the file path or another one

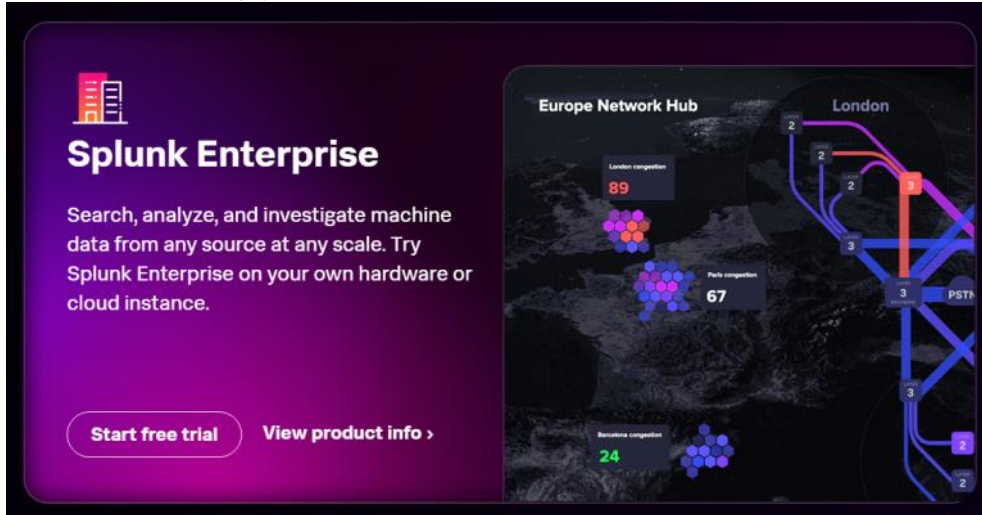
```
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ jq '[] | select(.event.PROCESS_ID == 3532) | .event.FILE_PATH' events.json
"C:\\Users\\tcm\\Downloads\\minikatz.exe"
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ jq '[] | select(.event.PROCESS_ID == 3532) | .event.HASH' events.json
"92804faaab2175dc501d73e814663058c78c0a042675a8937266357bcfb96c50"
tcm@SOC101-ubuntu:~/Desktop/04_Siem/01_Log_Analysis$ jq '[] | select(.event.PROCESS_ID == 3532) | .event.PROCESS_ID' events.json
3532
```


Introduction to Splunk

Tuesday, April 21, 2026 11:16 AM

Installation of splunk:

- 1- Splunk.com
[Free Trials & Downloads | Splunk](#)



- 2- Go to splunk enterprise trail and make an account



- 3- Verify the email then choose the OS

Splunk Enterprise 9.2.2

Index 500 MB/Day. Sign up and download now. After 60 days you can convert to a perpetual free license or purchase a Splunk Enterprise license to continue using the expanded functionality designed for enterprise-scale deployments.

Choose Your Installation Package

Windows	Linux	Mac OS
64-bit	3.x+, 4.x+, or 5.4.x kernel Linux distributions	
	.tgz	684.6 MB
		Download Now
		Copy wget link
		More v
	.deb	521.47 MB
		Download Now
		Copy wget link
		More v
	.rpm	684.44 MB
		Download Now
		Copy wget link
		More v

Am using linux but u can use other

- 4- The best practice is to download it in the opt file

```
tcm@SOC101-ubuntu:~$ cd Downloads/
tcm@SOC101-ubuntu:~/Downloads$ ls
splunk-9.2.2-d76edf6f0a15-Linux-x86_64.tgz  thunderbird.tmp
tcm@SOC101-ubuntu:~/Downloads$ sudo tar xvfz splunk-9.2.2-d76edf6f0a15-Linux-x86_64.tgz -C /opt
```

This will take up to 5 minutes to complete

- 5- Go to opt/splunk/bin and start splunk

Add username and password

```
tcm@SOC101-ubuntu:/opt/splunk/bin$ sudo ./splunk start --accept-license
```

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherwise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

```
Please enter an administrator username: tcm
Password must contain at least:
* 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password:
```

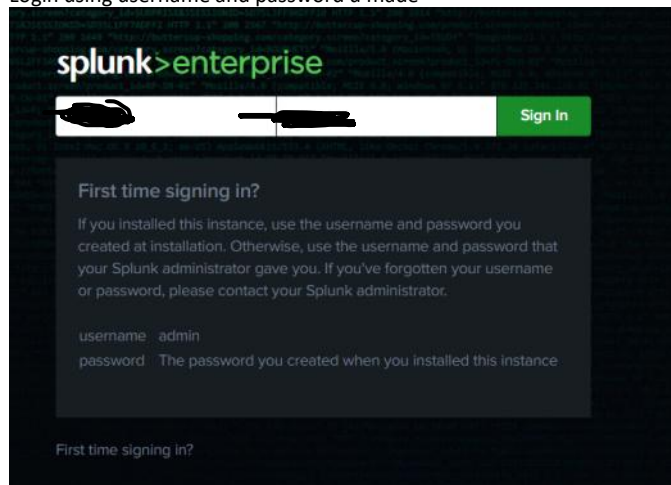
Go to localhost at the port

```
Waiting for web server at http://127.0.0.1:8000 to be available..... Done
```

If you get stuck, we're here to help.
Look for answers here: <http://docs.splunk.com>

The Splunk web interface is at <http://SOC101-ubuntu:8000>

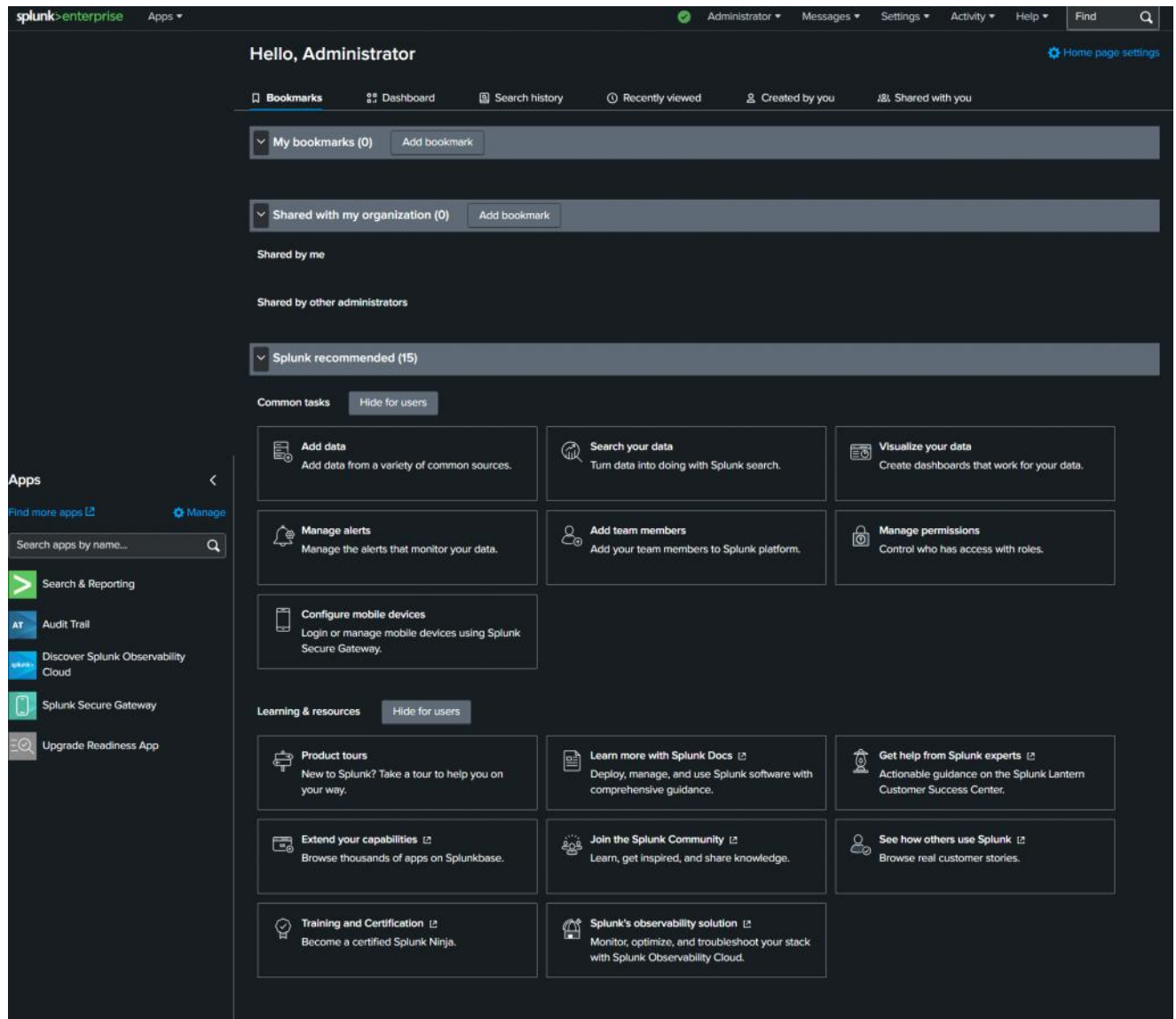
- 6- Login using username and password u made



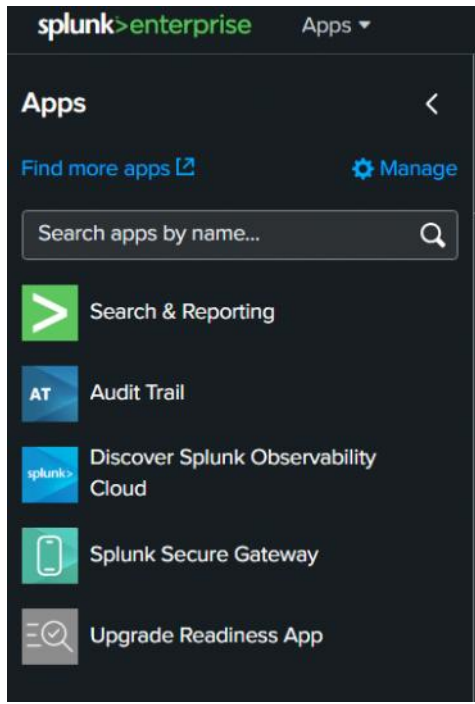
- 7- To make it auto start and see the statuses

```
tcm@SOC101-ubuntu:/opt/splunk/bin$ sudo ./splunk enable boot-start
Init script installed at /etc/init.d/splunk.
Init script is configured to run at boot.
tcm@SOC101-ubuntu:/opt/splunk/bin$ sudo ./splunk status
splunkd is running (PID: 1834).
splunk helpers are running (PIDs: 1845 3073 3200 3395).
```

8. Home page

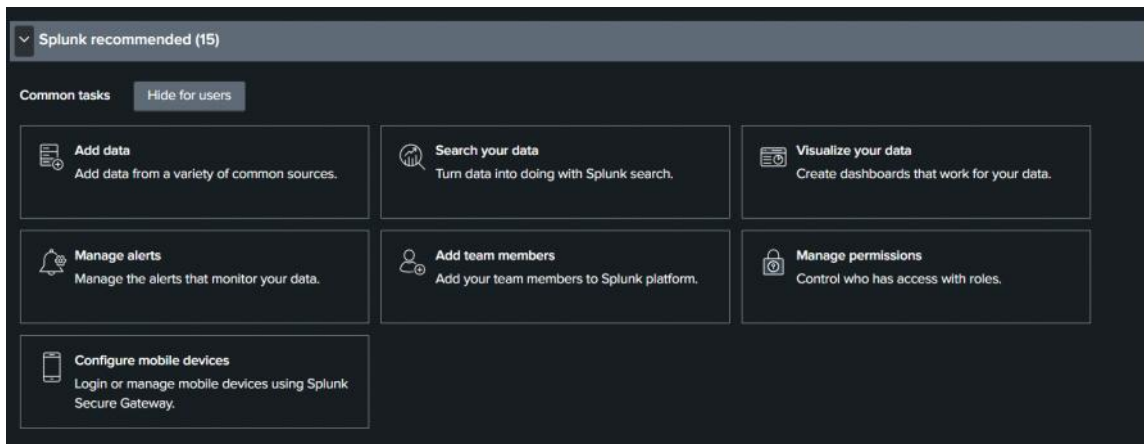


9. Apps (extensions)



These are extensions that we can use in Splunk but we will focus on search and reporting

10.



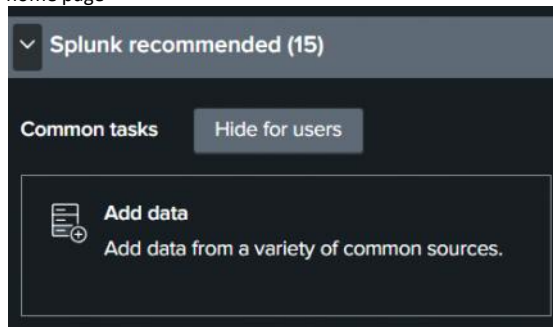
In this area we have the most used . Where we can add data manually or using forwarders , view data in different ways ,and adding permissions or users

Splunk Importing and Exploring Events

Wednesday, April 22, 2026 1:09 PM

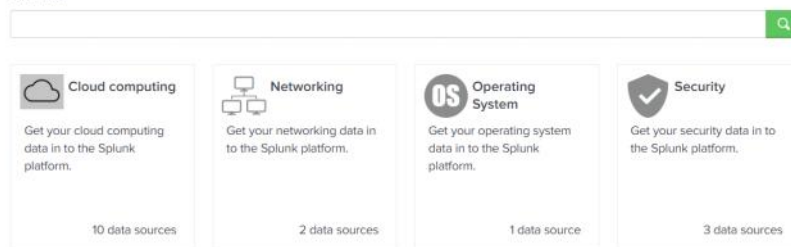
The first thing u need to start working with splunk is dataset

1. Go to add data in home page



Follow guides for onboarding popular data sources

Search bar



4 data sources in total

Or get data in with the following methods



And here we see methods to add data we are using the simple way upload a file

2. Make a notepad file and save it as data.csv with these data (or use online dataset)

```
timestamp,username,ip_address,status,event_type
2026-04-21 08:15:23,jdoe,192.168.1.10,success,login_success
2026-04-21 08:17:45,asmith,192.168.1.15,failed,login_failed
2026-04-21 08:18:12,jdoe,192.168.1.10,success,login_success
2026-04-21 08:19:30,mbrown,192.168.1.22,failed,login_failed
2026-04-21 08:20:05,mbrown,192.168.1.22,failed,login_failed
2026-04-21 08:20:45,mbrown,192.168.1.22,failed,login_failed
2026-04-21 08:21:10,mbrown,192.168.1.22,success,login_success
2026-04-21 08:22:33,asmith,192.168.1.15,failed,login_failed
2026-04-21 08:23:01,asmith,192.168.1.15,success,login_success
2026-04-21 08:24:55,jdoe,192.168.1.10,success,login_success
2026-04-21 08:25:17,guest,10.0.0.5,failed,login_failed
2026-04-21 08:25:45,guest,10.0.0.5,failed,login_failed
2026-04-21 08:26:12,guest,10.0.0.5,failed,login_failed
2026-04-21 08:26:50,guest,10.0.0.5,failed,login_failed
2026-04-21 08:27:30,guest,10.0.0.5,success,login_success
2026-04-21 08:28:11,admin,203.0.113.42,failed,login_failed
2026-04-21 08:28:35,admin,203.0.113.42,failed,login_failed
2026-04-21 08:29:02,admin,203.0.113.42,failed,login_failed
2026-04-21 08:29:40,admin,203.0.113.42,failed,login_failed
2026-04-21 08:30:10,admin,203.0.113.42,failed,login_failed
```

```

2026-04-21 08:30:45,admin,203.0.113.42,success,login_success
2026-04-21 08:31:20,jdoe,192.168.1.10,success,login_success
2026-04-21 08:32:05,asmith,192.168.1.15,failed,login_failed
2026-04-21 08:32:40,asmith,192.168.1.15,failed,login_failed
2026-04-21 08:33:15,asmith,192.168.1.15,success,login_success
2026-04-21 08:34:01,mbrown,192.168.1.22,success,login_success
2026-04-21 08:35:22,guest,10.0.0.5,failed,login_failed
2026-04-21 08:36:10,guest,10.0.0.5,failed,login_failed
2026-04-21 08:36:55,guest,10.0.0.5,success,login_success

```

```

data.csv
File Edit View

timestamp,username,ip_address,status,event_type
2026-04-21 08:15:23,jdoe,192.168.1.10,success,login_success
2026-04-21 08:17:45,asmith,192.168.1.15,failed,login_failed
2026-04-21 08:18:12,jdoe,192.168.1.10,success,login_success
2026-04-21 08:19:30,mbrown,192.168.1.22,failed,login_failed
2026-04-21 08:20:05,mbrown,192.168.1.22,failed,login_failed
2026-04-21 08:20:45,mbrown,192.168.1.22,failed,login_failed
2026-04-21 08:21:10,mbrown,192.168.1.22,success,login_success
2026-04-21 08:22:33,asmith,192.168.1.15,failed,login_failed
2026-04-21 08:23:01,asmith,192.168.1.15,success,login_success
2026-04-21 08:24:55,jdoe,192.168.1.10,success,login_success
2026-04-21 08:25:17,guest,10.0.0.5,failed,login_failed
2026-04-21 08:25:45,guest,10.0.0.5,failed,login_failed
2026-04-21 08:26:12,guest,10.0.0.5,failed,login_failed
2026-04-21 08:26:50,guest,10.0.0.5,failed,login_failed
2026-04-21 08:27:30,guest,10.0.0.5,success,login_success
2026-04-21 08:28:11,admin,203.0.113.42,failed,login_failed
2026-04-21 08:28:35,admin,203.0.113.42,failed,login_failed
2026-04-21 08:29:02,admin,203.0.113.42,failed,login_failed
2026-04-21 08:29:40,admin,203.0.113.42,failed,login_failed
2026-04-21 08:30:10,admin,203.0.113.42,failed,login_failed
2026-04-21 08:30:45,admin,203.0.113.42,success,login_success
2026-04-21 08:31:20,jdoe,192.168.1.10,success,login_success
2026-04-21 08:32:05,asmith,192.168.1.15,failed,login_failed
2026-04-21 08:32:40,asmith,192.168.1.15,failed,login_failed
2026-04-21 08:33:15,asmith,192.168.1.15,success,login_success
2026-04-21 08:34:01,mbrown,192.168.1.22,success,login_success
2026-04-21 08:35:22,guest,10.0.0.5,failed,login_failed
2026-04-21 08:36:10,guest,10.0.0.5,failed,login_failed
2026-04-21 08:36:55,guest,10.0.0.5,success,login_success

```

3. Use the upload method and add the file

Select Source

Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. [Learn More](#)

Selected File: **data.csv**

Select File

Drop your data file here

The maximum file upload size is 500 Mb

File Successfully Uploaded

4. Press next and go to input setting

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

Host field value:

☒ Constant value
☐ Regular expression on path
☐ Segment in path

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index: [Create a new index](#)

Here we need input the file of the machine we took the log file from and create a new index and just add a name for it and save

New Index

General Settings

Index Name:
Set index name (e.g., INDEX_NAME). Search using index=INDEX_NAME.

Index Data Type: ☒ Events ☐ Metrics
The type of data to store (event-based or metrics).

Home Path:
Hot/warm db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/db).

Cold Path:
Cold db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/colddb).

Thawed Path:
Thawed/resurrected db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/thaweddb).

Data Integrity Check: ☒ Enable ☐ Disable
Enable this if you want Splunk to compute hashes on every slice of your data for the purpose of data integrity.

Max Size of Entire Index:
Maximum target size of entire index.

Save **Cancel**

The index is used to make more than one log file so we can call each one alone if there is login file and system log file and network log file I can call each one alone if they are from same host

After u add the index press next until submit and go to home page and press search and report

- 1-make the time range all 2-in search bar type index

New Search

Index="data_login"

Time range: All time

29 events (before 4/22/26 1:58:26.000 PM) No Event Sampling

Events (29) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

Selected Fields: host, source, sourcetype

Interesting Fields: date_hour, date_mday, date_minute, date_month, date_second

Time	Event
4/21/26 8:36:55.000 AM	2026-04-21 08:36:55, guest, 10.0.0.5, success, login_success host = SOC-m8a8a8y source = data.csv sourcetype = csv
4/21/26 8:36:10.000 AM	2026-04-21 08:36:10, guest, 10.0.0.5, failed, login_failed host = SOC-m8a8a8y source = data.csv sourcetype = csv
4/21/26 8:35:22.000 AM	2026-04-21 08:35:22, guest, 10.0.0.5, failed, login_failed host = SOC-m8a8a8y source = data.csv sourcetype = csv
4/21/26 8:34:01.000 AM	2026-04-21 08:34:01, mbrown, 192.168.1.22, success, login_success host = SOC-m8a8a8y source = data.csv sourcetype = csv

In search bar u can search using - host , index , source - and when u are typing in search bar u have a helper showing u options

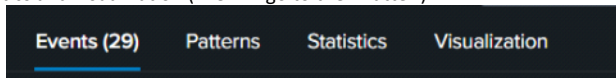
Note lets walkthrough the dashboard of search

1. Here we see the number of events

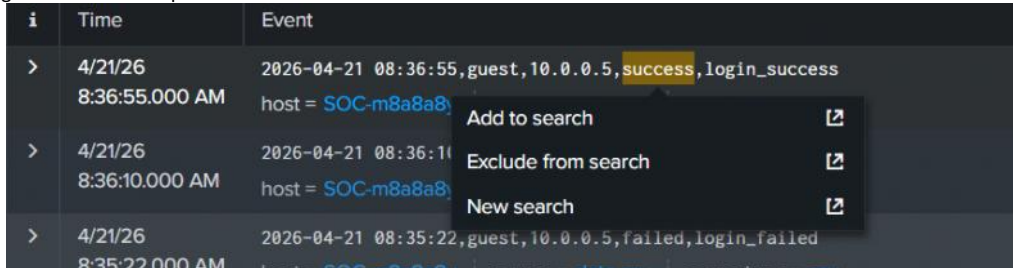
29 events (before 4/22/26 2:18:38.000 PM)

- Event tab : we see all events

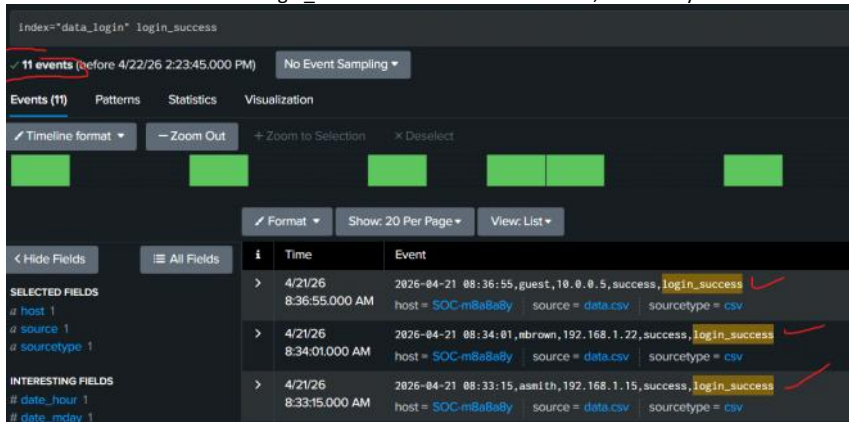
Pattern : is used to show patterns like multi failed login (comes handy when having many events)
Statistics and visualization (we will go to them latter)



- If we go to an event and press on it we can add it to search bar

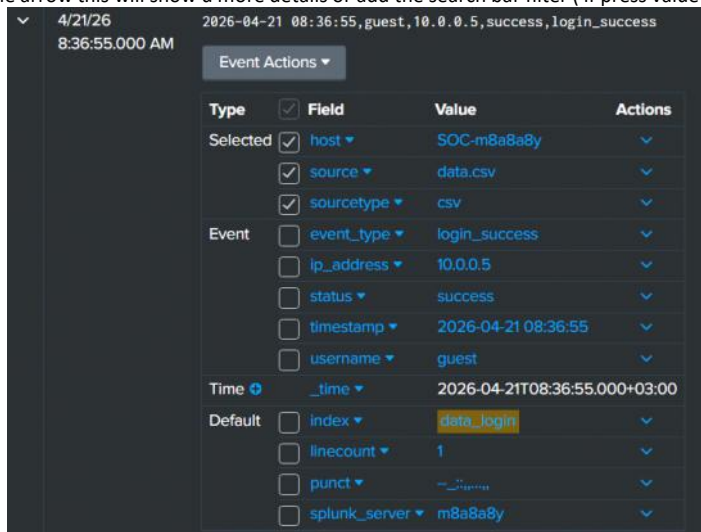


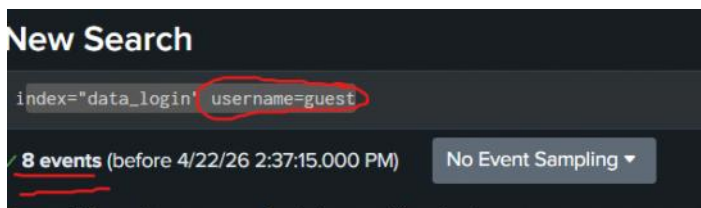
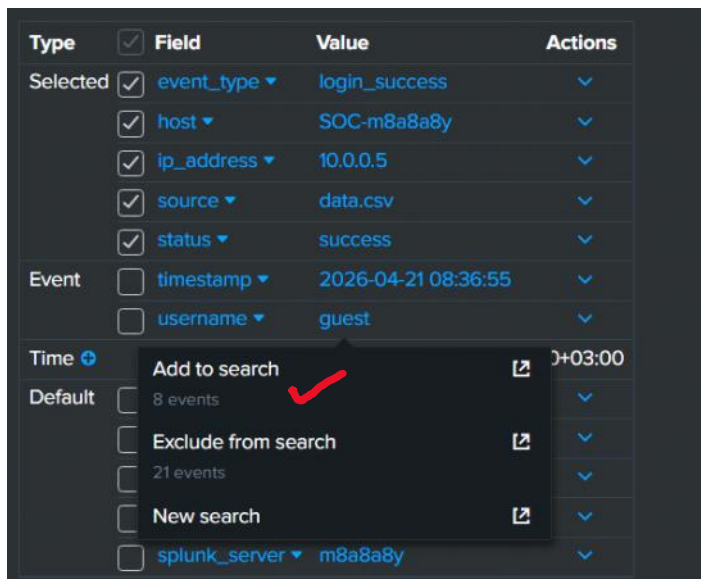
All the events shown are the login_success other than that are not , that's why the number of events are less



- More details

If u press the arrow this will show u more details or add the search bar filter (if press value)

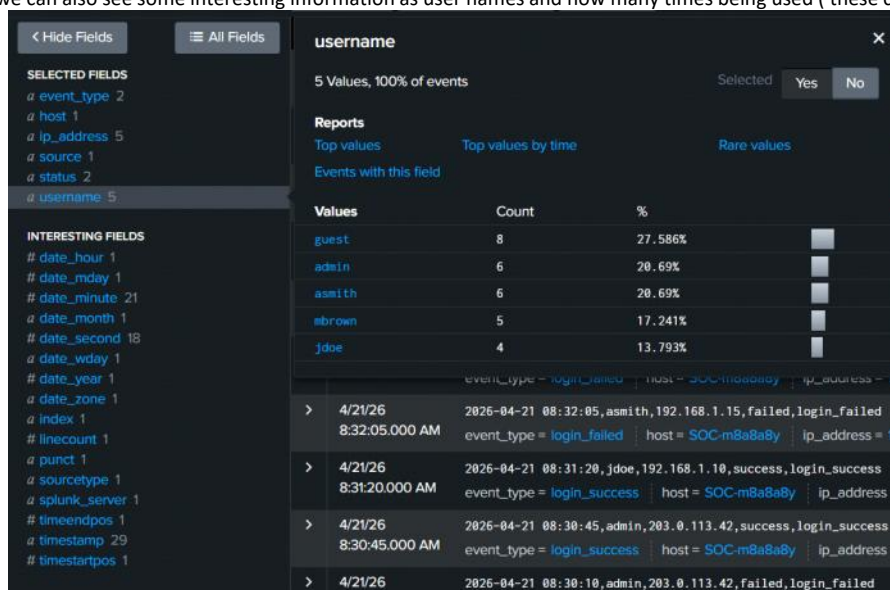




5. Fields

Here we can see more fields we can add to the event if select yes

And we can also see some interesting information as user names and how many times being used (these can be usefull to see the most ip or port been used)



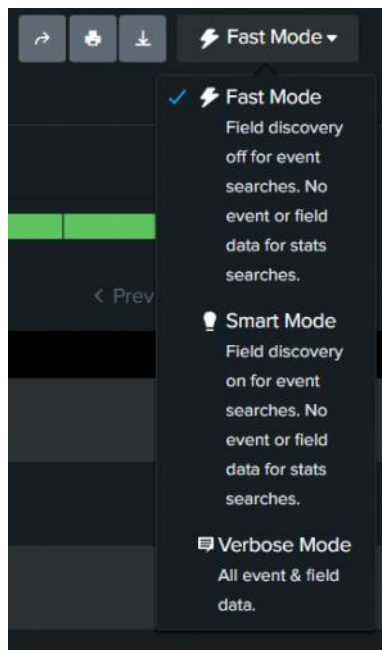
6. Others

Modes :we have 3 mods each one with details

Share : we can share the link

Print : we can have a PDF of it

Download: can download the file as CSV or JSON or XML



Splunk Search Processing Language (SPL)

Wednesday, April 22, 2026 2:46 PM

Is a search processing language used by Splunk for filtering the events , but it not the only one other SIEM solutions use other languages (example elastic use the Kibana)

1- Keywords : it's the normal way by entering a word in search bar

The screenshot shows the Splunk search interface with the query `index="sample_index" failed`. The search results are displayed in a table format. The left sidebar shows the search bar and the search results summary. The main table has columns for Time and Event. The search results show four events, all of which are failed login attempts.

Time	Event
7/20/24 12:55:00.000 PM	2024-07-20 12:55:00,jdoe,login,192.168.1.101,failed
7/20/24 12:35:00.000 PM	2024-07-20 12:35:00,bwilliams,login,192.168.1.105,failed
7/20/24 12:15:00.000 PM	2024-07-20 12:15:00,mjohnson,login,192.168.1.103,failed
7/20/24 12:10:00.000 PM	2024-07-20 12:10:00,jdoe,login,192.168.1.101,failed

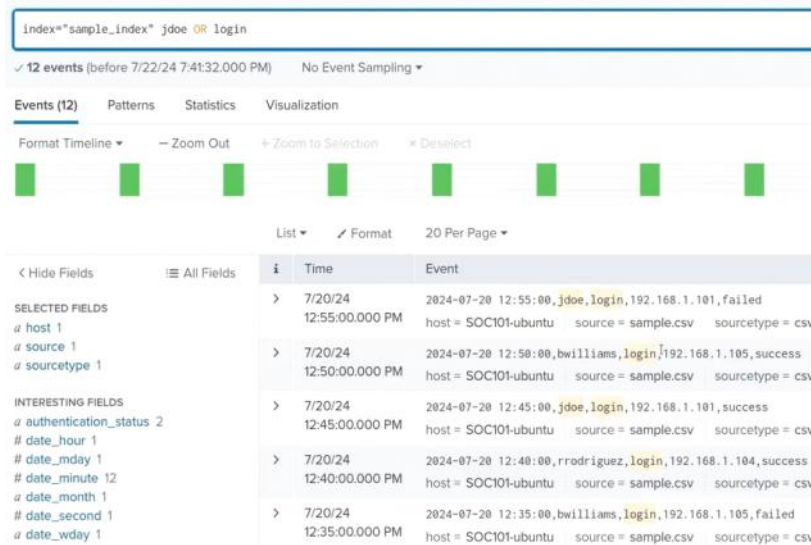
2- Bolian

AND : all should be there to get back a result

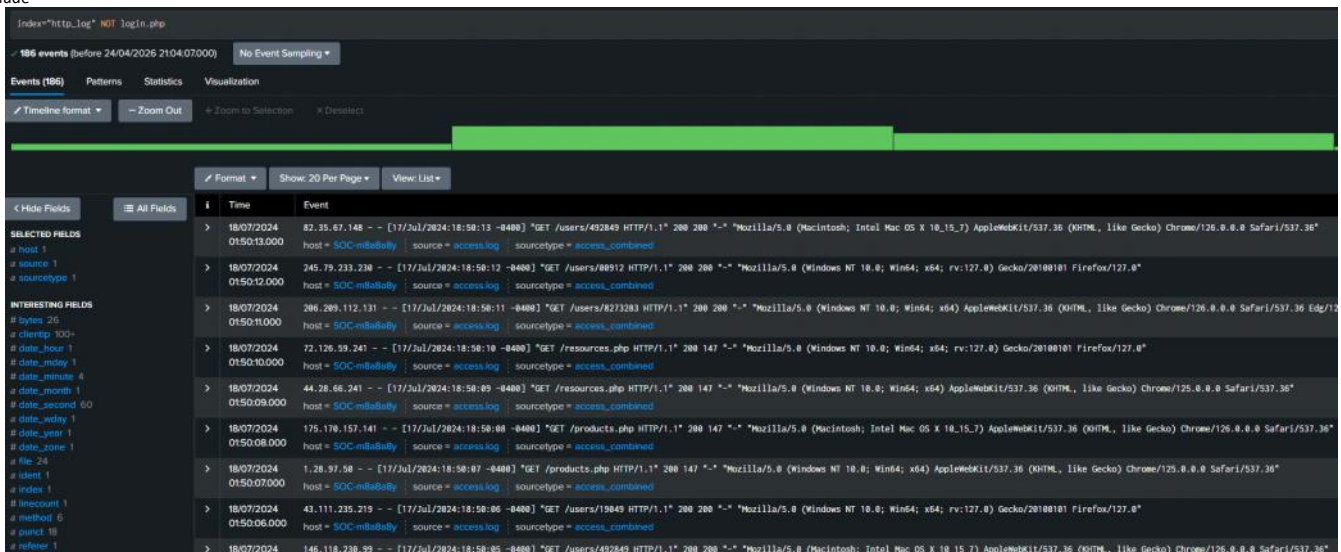
The screenshot shows the Splunk search interface with the query `index="sample_index" jdoe AND login`. The search results are displayed in a table format. The left sidebar shows the search bar and the search results summary. The main table has columns for Time and Event. The search results show four events, all of which are login attempts by the user 'jdoe'.

Time	Event
7/20/24 12:55:00.000 PM	2024-07-20 12:55:00,jdoe,login,192.168.1.101,failed
7/20/24 12:45:00.000 PM	2024-07-20 12:45:00,jdoe,login,192.168.1.101,success
7/20/24 12:10:00.000 PM	2024-07-20 12:10:00,jdoe,login,192.168.1.101,failed
7/20/24 12:00:00.000 PM	2024-07-20 12:00:00,jdoe,login,192.168.1.101,success

OR: one should be there to show a result



NOT : to exclude



And u can make a combination between them but u have the order will control the filtering
Example:

`index="http_sample" AND NOT method=GET OR method=POST`
This will filter on method=GET and then this result will be filtered with method=POST

`index="http_sample" AND NOT (method=GET OR method=POST)`
This will filter between the (...) first then this result will be filtered with AND NOT

3- Wildcard (*) : make better matching

For example if search only for fail u will get the word fail but fail* will show anywhere u see fail, failed

index="sample_index" failed

✓ 4 events (before 7/22/24 7:42:54.000 PM) No Event Sampling

Events (4) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

Hide Fields All Fields

SELECTED FIELDS

- # host 1
- # source 1
- # sourcetype 1

INTERESTING FIELDS

- # authentication_status 1
- # date_hour 1
- # date_mday 1
- # date_minute 4
- # date_month 1

i	Time	Event
>	7/20/24 12:55:00.000 PM	2024-07-20 12:55:00,jdoe,login,192.168.1.101,failed host = SOC101-ubuntu source = sample.csv sourcetype = csv
>	7/20/24 12:35:00.000 PM	2024-07-20 12:35:00,bwilliams,login,192.168.1.105,failed host = SOC101-ubuntu source = sample.csv sourcetype = csv
>	7/20/24 12:15:00.000 PM	2024-07-20 12:15:00,mjohnson,login,192.168.1.103,failed host = SOC101-ubuntu source = sample.csv sourcetype = csv
>	7/20/24 12:10:00.000 PM	2024-07-20 12:10:00,jdoe,login,192.168.1.101,failed host = SOC101-ubuntu source = sample.csv sourcetype = csv

U can use it anywhere like in time

New Search

index="sample_index" 12:8:00

✓ 12 events (before 7/22/24 7:44:39.000 PM) No Event Sampling

Events (12) Patterns Statistics Visualization

Format Timeline Zoom Out Zoom to Selection Deselect

Hide Fields All Fields

SELECTED FIELDS

- # host 1
- # source 1
- # sourcetype 1

INTERESTING FIELDS

- # authentication_status 2
- # date_hour 1
- # date_mday 1
- # date_minute 12
- # date_month 1
- # date_second 1
- # date_wday 1

i	Time	Event
>	7/20/24 12:55:00.000 PM	2024-07-20 12:55:00,jdoe,login,192.168.1.101,failed host = SOC101-ubuntu source = sample.csv sourcetype = csv
>	7/20/24 12:50:00.000 PM	2024-07-20 12:50:00,bwilliams,login,192.168.1.105,success host = SOC101-ubuntu source = sample.csv sourcetype = csv
>	7/20/24 12:45:00.000 PM	2024-07-20 12:45:00,jdoe,login,192.168.1.101,success host = SOC101-ubuntu source = sample.csv sourcetype = csv
>	7/20/24 12:40:00.000 PM	2024-07-20 12:40:00,rrodriguez,login,192.168.1.104,success host = SOC101-ubuntu source = sample.csv sourcetype = csv
>	7/20/24 12:35:00.000 PM	2024-07-20 12:35:00,bwilliams,login,192.168.1.105,failed host = SOC101-ubuntu source = sample.csv sourcetype = csv

4- Interesting Fields

Use the access.log in https://github.com/MalwareCube/SOC101/blob/main/course_files/04_SIEM.zip (password : LCTy2JmHQLB3uc9VxjeohENr)

Add to splunk and make a new index for it

The intrsting fields are fields by splunk that give it to show the top values in events

INTERESTING FIELDS

```
# bytes 30
# clientip 100+
# date_hour 1
# date_mday 1
# date_minute 4
# date_month 1
# date_second 60
# date_wday 1
# date_year 1
# date_zone 1
# file 25
# ident 1
# index 1
# linecount 1
# method 6
# punct 18
# referer 1
# req_time 100+
# splunk_server 1
# status 3
# timeendpos 7
# timestartpos 7
# uri 30
# uri_path 26
# user 1
# useragent 8
# version 2
```

The top values can show some interesting parts to filter one

INTERESTING FIELDS

```
# bytes 30
# clientip 100+
# date_hour 1
# date_mday 1
# date_minute 4
# date_month 1
# date_second 60
# date_wday 1
# date_year 1
# date_zone 1
# file 25
# ident 1
# index 1
# linecount 1
# method 6
# punct 18
# referer 1
# req_time 100+
# splunk_server 1
# status 3
# timeendpos 7
# timestartpos 7
```

file

25 Values, 99.157% of events

Selected

Reports

Top values Top values by time Rare values

Events with this field

Top 10 Values	Count	%
login.php	2,900	94.771%
admin.php	17	0.556%
resources.php	15	0.49%
02939232	14	0.458%
about.php	12	0.392%
912839213	11	0.359%
contact.php	11	0.359%
index.php	10	0.327%
00912	9	0.294%
492849	9	0.294%

If press on one of the values can be used to filter on it using the =

New Search

index="http_log" file="login.php"

Event

```
195.154.25.45 - - [17/Jul/2024:18:50:03 -0400] "GET /login.php HTTP/1.1" 200 782 "-" Mozilla/5.0 (Windows NT
host = SOC-m8a8a8y | source = access.log | sourcetype = access_combined

212.40.226.179 - - [17/Jul/2024:18:49:48 -0400] "GET /login.php HTTP/1.1" 200 782 "-" Mozilla/5.0 (Macintosh
host = SOC-m8a8a8y | source = access.log | sourcetype = access_combined

150.145.31.170 - - [17/Jul/2024:18:49:27 -0400] "GET /login.php HTTP/1.1" 200 782 "-" Mozilla/5.0 (Windows NT
host = SOC-m8a8a8y | source = access.log | sourcetype = access_combined

180.93.216.251 - - [17/Jul/2024:18:49:03 -0400] "GET /login.php HTTP/1.1" 200 782 "-" Mozilla/5.0 (Macintosh
host = SOC-m8a8a8y | source = access.log | sourcetype = access_combined

62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "POST /login.php HTTP/1.0" 200 312 "-" Mozilla/5.0 (Hydra)"
host = SOC-m8a8a8y | source = access.log | sourcetype = access_combined

62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "POST /login.php HTTP/1.0" 200 312 "-" Mozilla/5.0 (Hydra)"
host = SOC-m8a8a8y | source = access.log | sourcetype = access_combined

62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "POST /login.php HTTP/1.0" 200 312 "-" Mozilla/5.0 (Hydra)"
host = SOC-m8a8a8y | source = access.log | sourcetype = access_combined

62.122.201.246 - - [17/Jul/2024:18:49:02 -0400] "POST /login.php HTTP/1.0" 200 312 "-" Mozilla/5.0 (Hydra)"
host = SOC-m8a8a8y | source = access.log | sourcetype = access_combined
```

If u want to show all but discard one use !=

New Search

index="http_log" file="login.php"

82.35.67.148	- -	[17/Jul/2024:18:50:13 -0400]	"GET /users/492849 HTTP/1.1"	200 200	-"	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/127.0.0.0 Safari/537.36
host = SOC-m8a8a8y	:	source = access_log	:	sourcetype = access_combined		
245.79.233.230	- -	[17/Jul/2024:18:50:12 -0400]	"GET /users/00912 HTTP/1.1"	200 200	-"	Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:127.0) Gecko/20100101 Firefox/127.0"
host = SOC-m8a8a8y	:	source = access_log	:	sourcetype = access_combined		
206.209.112.131	- -	[17/Jul/2024:18:50:11 -0400]	"GET /users/8273283 HTTP/1.1"	200 200	-"	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/127.0.0.0 Safari/537.36
host = SOC-m8a8a8y	:	source = access_log	:	sourcetype = access_combined		
72.126.59.241	- -	[17/Jul/2024:18:50:10 -0400]	"GET /resources.php HTTP/1.1"	200 147	-"	Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:127.0) Gecko/20100101 Firefox/127.0"
host = SOC-m8a8a8y	:	source = access_log	:	sourcetype = access_combined		
44.28.66.241	- -	[17/Jul/2024:18:50:09 -0400]	"GET /resources.php HTTP/1.1"	200 147	-"	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/127.0.0.0 Safari/537.36
host = SOC-m8a8a8y	:	source = access_log	:	sourcetype = access_combined		
175.170.157.141	- -	[17/Jul/2024:18:50:08 -0400]	"GET /products.php HTTP/1.1"	200 147	-"	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/127.0.0.0 Safari/537.36
host = SOC-m8a8a8y	:	source = access_log	:	sourcetype = access_combined		
1.28.97.50	- -	[17/Jul/2024:18:50:07 -0400]	"GET /products.php HTTP/1.1"	200 147	-"	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/127.0.0.0 Safari/537.36
host = SOC-m8a8a8y	:	source = access_log	:	sourcetype = access_combined		
43.111.235.219	- -	[17/Jul/2024:18:50:06 -0400]	"GET /users/19049 HTTP/1.1"	200 200	-"	Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:127.0) Gecko/20100101 Firefox/127.0"
host = SOC-m8a8a8y	:	source = access_log	:	sourcetype = access_combined		
146.118.230.99	- -	[17/Jul/2024:18:50:05 -0400]	"GET /users/492849 HTTP/1.1"	200 200	-"	Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/127.0.0.0 Safari/537.36

In some cases like the status code u can use less than or more than : status>200 or as status<200
And u have >= or <=

Splunk Search Commands

Sunday, April 26, 2026 12:18 PM

After we learned about SPL we are moving to advanced

We are going to use the pip or "|" to make the usage of statistics

1- Stats

Has many usage the most popular is counting by as parameter

Search

index="access_log" | stats count by

No Event Sampling ▾

> SPL Search History ?

How to Search

If you are not familiar with the search

stats count by action, host

stats count by clientip

stats count by clientip

stats count by host

stats count by src_ip,dest_ip,dest_port

index="access_log" | stats count by clientip

index="access_log" | stats count by clientip

index="access_log" | stats count by host

stats

Provides statistics, grouped optionally by field.

Example:

```
sourcetype=access_combined | top limit=100 referer_domain | stats sum(count)
```

Command History

Command History

Command History

Command History

Command History

Matching Search

Matching Search

Matching Search

[Learn More](#)

Example count the IPs

index="access_log" | stats count by clientip

✓ 3,086 events (before 4/26/26 12:26:34.000 PM) No Event Sampling ▾

Events Patterns **Statistics (161)** Visualization

Show: 100 Per Page ▾ Format ▾ Preview: On

clientip ↕

1.28.97.50

10.165.246.243

10.94.48.215

100.131.156.16

100.43.255.39

100.54.82.53

104.228.114.80

106.192.94.135

108.175.99.111

11.179.153.98

11.195.89.153

111.93.75.121

Note : clientip is a field from the event

i	Time	Event
✓	7/18/24 1:50:13.000 AM	82.35.67.148 - - [17/Jul/2024:18:50:13 -0400] "GET /users/4927.36 (KHTML, like Gecko) Chrome/126.0.0.0 Safari/537.36"
Event Actions ▼		
Type	Field	Value
Selected	host	SOC-m8a8a8y
	source	access.log
Time	_time	2024-07-18T01:50:13.000+03:00
Default	index	access_log
	linecount	1
	sourcetype	access_combined
	splunk_server	m8a8a8y

2- Sort

U can use sort alone or u can use it after a previous pip using the output of the current command in this case using client ip and count

clientip	count
1.28.97.50	1
10.165.246.243	1
10.94.48.215	1
100.131.156.16	1
100.43.255.39	1
100.54.82.53	1
104.228.114.80	1
106.192.94.135	1
108.175.99.111	7
11.179.153.98	1

Result : the (- count : means descending order , count : means ascending order)

New Search

Save As

Create Table View

Close

index="access_log" | stats count by clientip | sort - count

Time range: All time

Job

Fast Mode

3,086 events (before 4/26/26 12:35:25.000 PM)

No Event Sampling

Events

Patterns

Statistics (161)

Visualization

Show: 100 Per Page

Format

Preview: On

< Prev

1

2

Next >

clientip	count
62.122.201.246	2926
108.175.99.111	7
1.28.97.50	1
10.165.246.243	1
10.94.48.215	1
100.131.156.16	1
100.43.255.39	1
100.54.82.53	1
104.228.114.80	1
106.192.94.135	1

3- Table

This is used to make a table using some fields needed usually for reports

index="access_log" | table _time,clientip,source

Time range: All time

✓ 3,086 events (before 4/26/26 12:40:36.000 PM) No Event Sampling

Events Patterns **Statistics (3,086)** Visualization

Show: 100 Per Page Format Preview: On

< Prev 1 2 3 4 5 6 7 8 ... Next >

_time	clientip	source
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log
2024-07-18 01:48:53	62.122.201.246	access.log

4- Dedup
Used to remove the duplicates

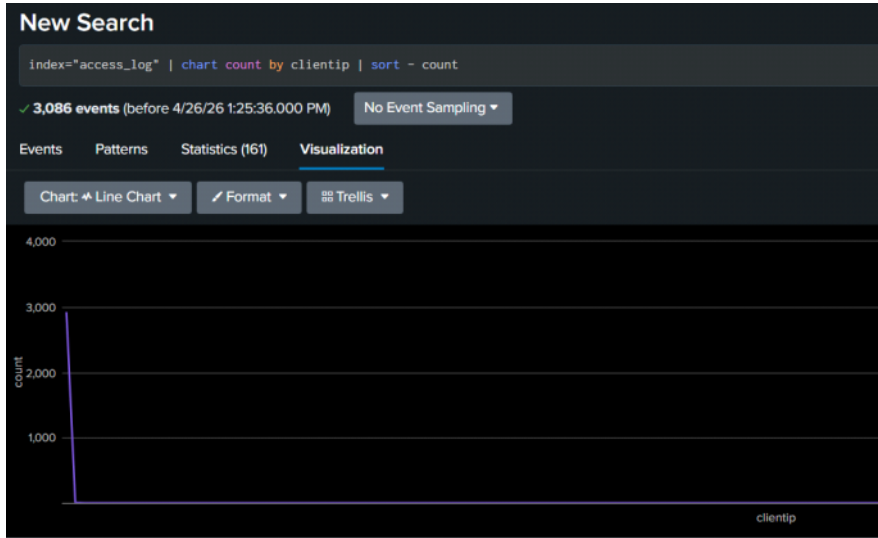
New Search

index="access_log" | table _time,clientip,source | dedup clientip

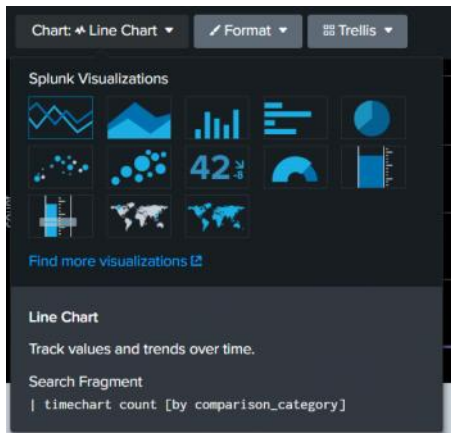
✓ 3,086 events (before 4/26/26 1:20:31.000 PM) No Event Sampling

Events Patterns **Statistics (161)** Visualization

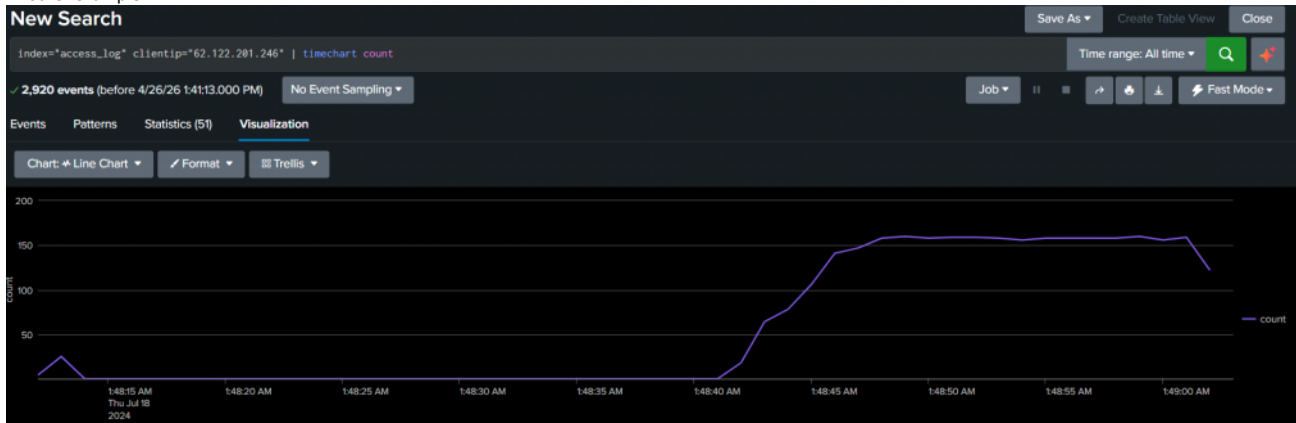
5- Chart
Is used to make visualization



and in Chart drop menu u can change the way off the chart



Another example



6- Search

Is used to match a pattern

New Search Save As

index="http_sample" clientip=62.122.201.246
 | table _time, clientip, useragent
 | search useragent="Nmap"

✓ 29 events (before 7/22/24 8:31:35.000 PM) No Event Sampling Job

Events Patterns **Statistics (29)** Visualization

20 Per Page Format Preview

_time ↕	clientip ↕	useragent ↕
2024-07-17 18:48:13	62.122.201.246	Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
2024-07-17 18:48:13	62.122.201.246	Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
2024-07-17 18:48:13	62.122.201.246	Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
2024-07-17 18:48:13	62.122.201.246	Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
2024-07-17 18:48:13	62.122.201.246	Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
2024-07-17 18:48:13	62.122.201.246	Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
2024-07-17 18:48:13	62.122.201.246	Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)
2024-07-17 18:48:13	62.122.201.246	Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)

7- Iplocation

Is used to know the location of the ip

New Search

index="access_log" | iplocation clientip

✓ 3,086 events (before 4/26/26 1:49:49.000 PM) No Event Sampling

Events (3,086) Patterns Statistics

Timeline format Zoom Out

Hide Fields All Fields

SELECTED FIELDS

- # host 1
- # source 1

INTERESTING FIELDS

- # City 100+
- # clientip 100+
- # Country 37
- # index 1
- # lat 100+
- # linecount 1
- # lon 100+
- # Region 77

Country

37 Values, 99.417% of events

Selected Yes No

Reports

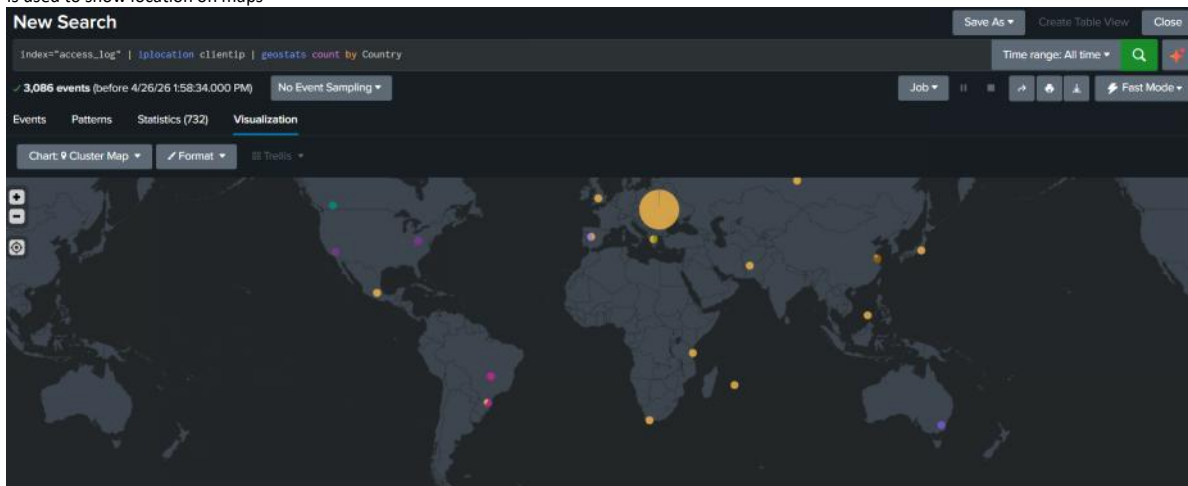
- Top values
- Top values by time
- Rare values

Events with this field

Top 10 Values	Count	%
Ukraine	2,920	95.176%
United States	67	2.184%
China	11	0.358%
Germany	7	0.228%
Canada	5	0.163%
France	5	0.163%
Japan	5	0.163%
Switzerland	4	0.13%
Australia	3	0.098%
Brazil	3	0.098%

8- Geostats

Is used to show location on maps

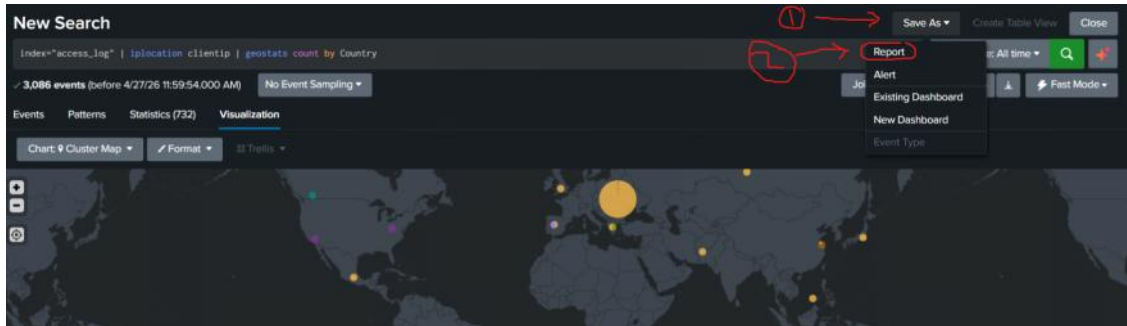


Splunk Reports and Alerts

Monday, April 27, 2026 11:52 AM

1- Reports

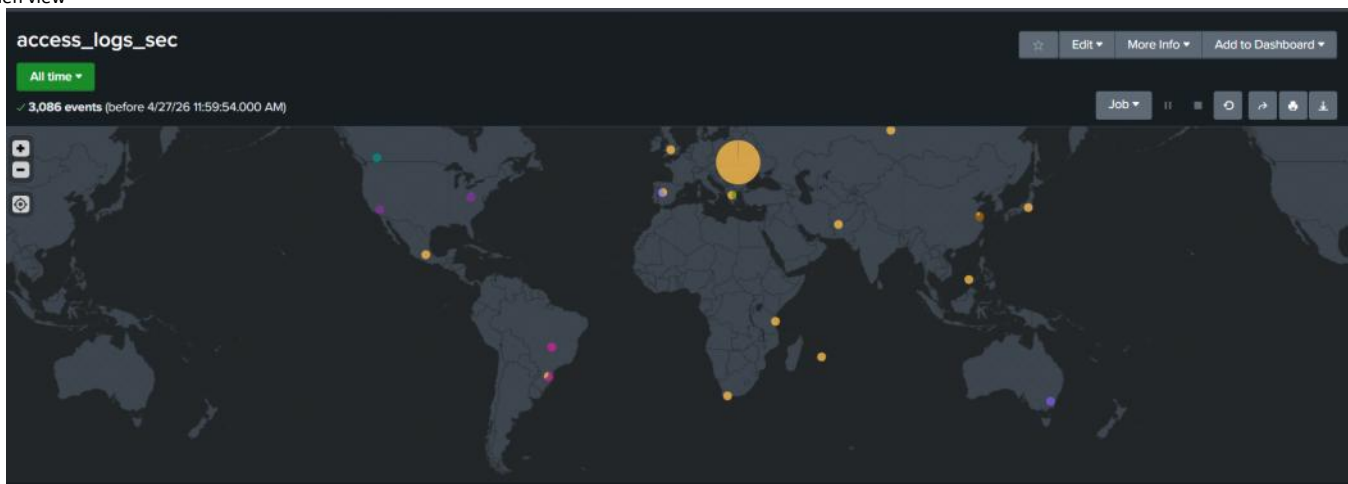
When we want to make report for and event in Splunk



Give it a name and details

The screenshot shows the 'Save As Report' dialog box. It has a title field, a description field with the text 'optional', and a content field with three icons: a location pin, a plus sign, and a calendar. Below these is a 'Time Range Picker' with 'Yes' and 'No' buttons. At the bottom right, there are 'Cancel' and 'Save' buttons.

Save and then view



2- Alert

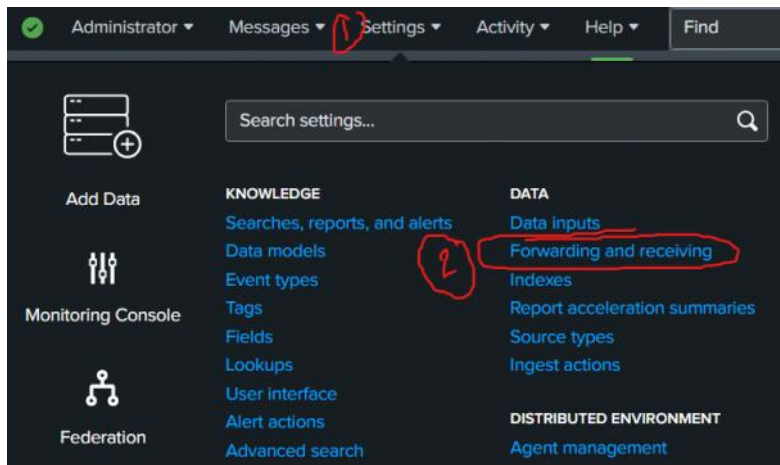
Is used to make a alert when the same search query appears

Example of ../../ (url encoded)

Splunk Deploying a Forwarder and Generating Real-Time Alerts

Monday, April 27, 2026 1:44 PM

All the before we a premade or taken logs files now we are making a Realtime Forwarder



Forward data
Set up forwarding between two or more Splunk instances.

Type	Actions
Forwarding defaults	
Configure forwarding	+ Add new

Receive data
Configure this instance to receive data forwarded from other instances.

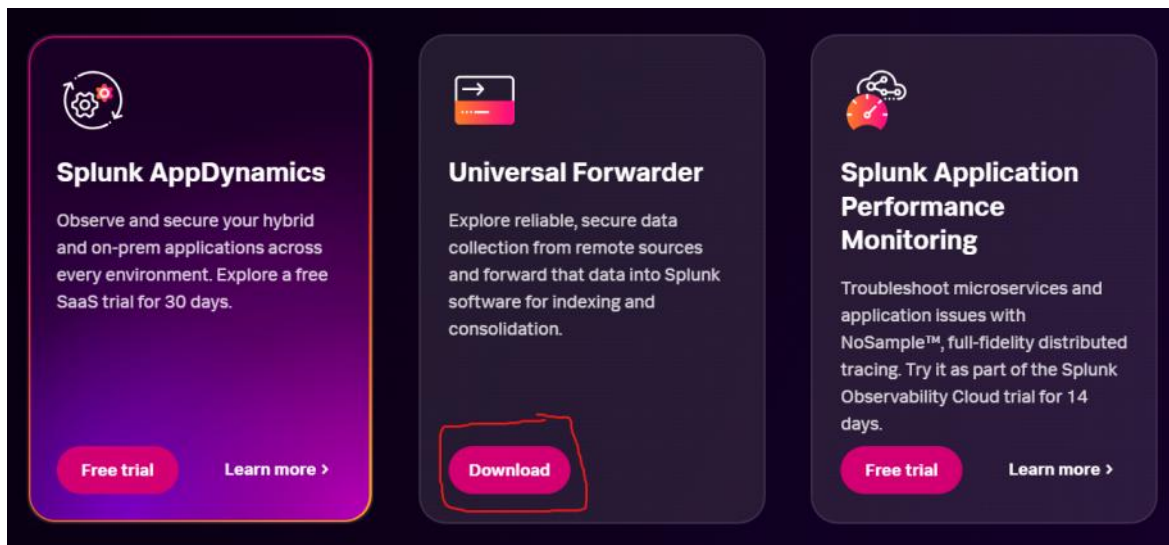
Type	Actions
Configure receiving	+ Add new

Configure receiving
Set up this Splunk instance to receive data from forwarder(s).

Listen on this port *

For example, 9997 will receive data on TCP port 9997.

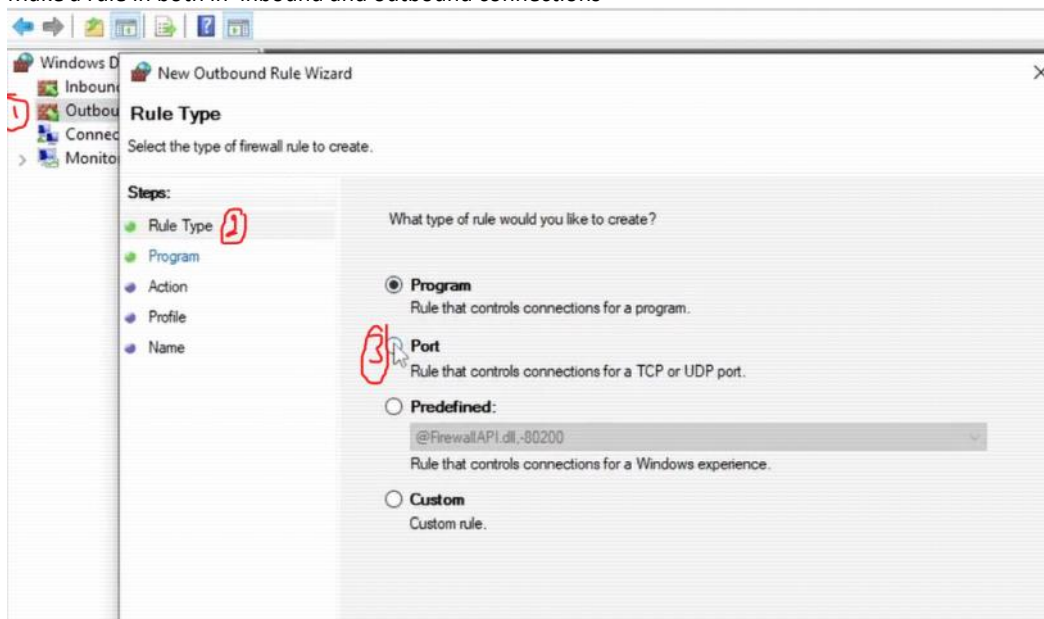
The port now is listening on the port we need to install Splunk universal forwarder : [Download Universal Forwarder for Remote Data Collection | Splunk](#)



Before I setup the forwarder go to the firewall and make a rule to make the logs pass



Make a rule in both in inbound and outbound connections



8089 and the listening port u make in splunk

Steps:

- Rule Type
- Protocol and Ports
- Action
- Profile
- Name

Does this rule apply to TCP or UDP?

☒ TCP
☐ UDP

Does this rule apply to all remote ports or specific remote ports?

☐ All remote ports
☒ Specific remote ports:

Example: 80, 443, 5000-5010

Steps:

- Rule Type
- Protocol and Ports
- Action
- Profile
- Name

What action should be taken when a connection matches the specified conditions?

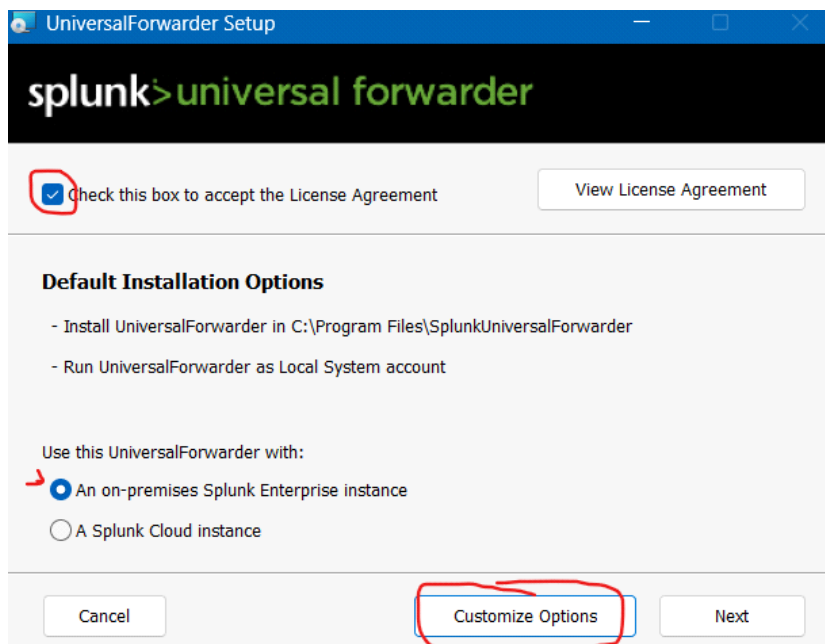
☒ **Allow the connection**
This includes connections that are protected with IPsec as well as those are not.

☐ **Allow the connection if it is secure**
This includes only connections that have been authenticated by using IPsec. Connections will be secured using the settings in IPsec properties and rules in the Connection Security Rule node.
[Customize...](#)

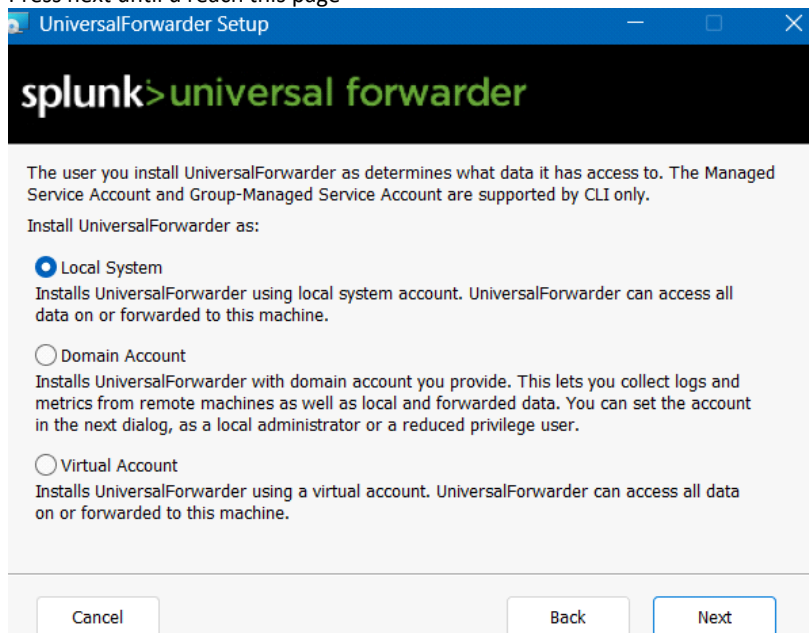
☐ **Block the connection**

Give it a name and make the 2nd one (inbound or outbound)

Now install the Splunk forwarder



Press next until u reach this page



Choose the logs u want to pass

UniversalForwarder Setup

splunk>universal forwarder

Windows Event Logs

☐ Application Logs

☒ Security Log

☒ System Log

☐ Forwarded Events Log

☐ Setup Log

Active Directory Monitoring

☐ Enable AD monitoring

Path to monitor

File... Directory...

Performance Monitor

☐ CPU Load

☐ Memory

☐ Disk Space

☐ Network Stats

Cancel Back Next

Make username and password (or make random)

UniversalForwarder Setup

splunk>universal forwarder

Create credentials for the administrator account. The password must contain, at a minimum, 8 printable ASCII characters.

Username:

☒ Generate random password

Password:

Confirm password:

Cancel Back Next

Enter the ip of the machine u are listening on and the port

UniversalForwarder Setup

splunk>universal forwarder

If you intend to use a Splunk deployment server to configure this UniversalForwarder, please specify the host or IP, and port (default port is 8089). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.

Deployment Server

Hostname or IP

:

Enter the hostname or IP of your deployment server, e.g. ds.splunk.com *default is 8089*

Cancel Back Next

Same IP but change the port

UniversalForwarder Setup

splunk>universal forwarder

If you intend to use a Splunk receiving indexer to configure this UniversalForwarder, please specify the host or IP, and port (default port is 9997). This is an optional step. However, UniversalForwarder needs either a deployment server or receiving indexer in order to do anything.

Receiving Indexer

Hostname or IP

:

Enter the hostname or IP of your receiving indexer, e.g. ds.splunk.com *default is 9997*

Cancel Back Next

When it finished go to search in Splunk and look in data summary

Data Summary

Hosts (1) Sources (2) Sourcetypes (2)

filter

Host	Count	Last Update
m8a8a8y	77,872	4/27/26 11:46:37000 PM

Documentation Tutorial **Data Summary**

U can now get it in search bar using host="hostname" and filter the events